Algebra Is Half the Battle: Verifying Presentations of Graded Unipotent Chevalley Groups

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Abstract

Graded unipotent Chevalley groups are an important family of groups on matrices with polynomial entries over a finite field. Using the Lean theorem prover, we verify that three such groups, namely, the A_3- and the two B_3-type groups, satisfy a useful group-theoretic condition. Specifically, these groups are defined by a set of equations called Steinberg relations, and we prove that a certain canonical "smaller" set of Steinberg relations suffices to derive the rest.

Our work is motivated by an application for building topologically-interesting objects called higher-dimensional expanders (HDXs). In the past decade, HDXs have formed the basis for many new results in theoretical computer science, such as in quantum error correction and in property testing. Yet despite the increasing prevalence of HDXs, only two methods of constructing them are known. One such method builds an HDX from groups that satisfy the aforementioned property, and the Chevalley groups we use are (essentially) the only ones currently known to satisfy it.

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Supplementary Material *Software*: `https://github.com/singerng/steinberg-formalization` archived at `swh:1:dir:dcac1d58f554132e469a8339afb6224cb72562db`

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1 Introduction

In this work, we verify recent results [25, 35] showing that a smaller-than-previously-known subset of the Steinberg relations are sufficient to define three specific Chevalley groups. As we explain later, these groups are used to construct objects with useful topological properties called higher-dimensional expanders (HDXs). While we do not formalize this connection to HDXs, we discuss how this connection motivated and shaped our work.

In order to even state the theorem we proved, we need to cover quite a bit of mathematical background (see Section 3). But informally, we verified the following:


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▶ **Theorem 1** ([25, 35]). *Let G be one of the three graded unipotent Chevalley groups we discuss in the paper. Then all of the Steinberg relations defining G can instead be derived from a certain subset of the relations, namely, from just codimension-1 relations, lifted relations, and definition relations. In other words, G is isomorphic to the group presented by these three kinds of relations.*

Our theorem verifies results from two papers [25,35], which we call "KOOS" based on the last names of the authors.[1] In our opinion, these papers were good candidates for computer verification: their proofs are filled with repetitive, mechanical, and precise calculations that are tedious for humans but rote for computers. The calculations in OS alone spanned 80 pages! Clearly, we thought, computer verification was needed to confirm these results.

Ideally, an automated theorem prover could do the work, but a supermajority of the proofs require solving group-rewriting problems, which are undecidable in general [33]. The natural next choice was to use an interactive theorem prover. We used Lean 4.

Lean [9] is an increasingly popular tool used to formalize mathematics and computer science. One of Lean's major selling points is the `mathlib` project [32], which formalizes many branches of math and CS. By placing key definitions, results, and proof tactics in the same project, mathematicians can build on each other's work. Our work depends on `mathlib`; through our work, we will contribute to it in turn.

Even though we made heavy use of Lean's tactics and automation, our proofs were still as mechanical as the ones in the KOOS papers. We tried to reduce some of the proof burden by using custom tactics and macros that automatically completed many lemmas for us, but ultimately, our proofs are a few thousand lines of symbol pushing. We hope to improve our custom automation in future work.

The rest of the paper is organized as follows. We first discuss the connection to HDXs that motivated our work (Section 2). We then define the Chevalley groups that we formalized (Section 3). Next, we give a tour of our formalization by discussing key definitions and the top-level theorem statement (Section 4), the overall shape of the proof (Section 5), and the automation we used (Section 6). We conclude with a reflection on our experience and a discussion of future work (Section 7).

You can find our proofs at `https://github.com/singerng/steinberg-formalization`.

2 Background on HDXs

While Theorem 1 is mathematically interesting on its own, KOOS (and we) were ultimately motivated by its corollary: a method of constructing topologically-useful objects called higher-dimensional expanders (HDXs). We did not formalize this corollary, but we discuss HDXs here to explain how our work fits into the broader research context.

HDXs have only been studied for about two decades, but they have already formed the basis for numerous results in theoretical CS, including better lower bounds for the Sum-of-Squares hierarchy [14, 21], new constructions for locally-testable error correction [13], and new algorithms for low-density quantum error correction [27] and property testing [3, 4, 12, 15]. HDXs also have interesting applications in higher-dimensional geometry [16–18, 23].

Interest in HDXs grew out of the study of graph expansion. An *expander graph* is a sparse graph with strong connectivity properties. Like HDXs, expander graphs are used in a wide range of applications, including error-correcting codes, entropy reduction for randomized algorithms, and the design of near-linear-time algorithms. Expander graphs have been studied for many decades; we refer the interested reader to a survey on the topic [20].

[1] "KO" is from the first paper, and "OS" is from the second. The "S" is the Singer of this paper.

Expansion can be defined in several different ways, but the various definitions fall under two umbrellas: *spectral* expansion and *topological* expansion. Informally, random walks over spectral expanders spread out quickly, while topological expanders do not contain any bottlenecks. In graph theory, these two kinds of expansion are closely related via the celebrated Cheeger inequality [1]. Thus, in order to build a topological expander, it is sufficient to build a spectral expander and then invoke this black-box relationship.

As their name suggests, *higher-dimensional expanders* (HDXs) generalize expansion for higher-dimensional objects [18, 29], such as simplicial complexes. But in this setting, spectral and topological expansion are *not* equivalent [19, 38], and proving topological expansion in particular can be difficult.

Broadly, there are only two known types of good HDX constructions: quotients of Bruhat-Tits buildings [5, 7, 28, 30, 31, 37]; and, more recently discovered, coset complexes of Chevalley groups [24–26, 35]. Both types are generally known to be spectral expanders, but only in certain special cases are they known to be topological expanders as well. In this paper, we focus on the topological properties of these Chevalley groups.

Chevalley groups were first used in HDX constructions by KO in 2018 [24]. Specifically, KO showed that the coset complexes of graded A_n-type Chevalley groups are spectral expanders. In their follow-up work [25], they showed that these complexes are also topological expanders by connecting topological expansion to group-theoretic properties of presentations (see Section 3.2). Separately, O'Donnell and Pratt [34] generalized KO's construction to show that coset complexes of graded Chevalley groups of other types (B_n, C_n, D_n, and exceptional) are spectral expanders. Most recently, OS [35] showed (again, using the expansion–group theory connection) that the B_3-type coset complexes are also topological expanders. Our work confirms the group-theoretic findings for the A_3- [25] and B_3- [35] types.

3 Chevalley groups, presentations, and our theorem (restated)

We now discuss Chevalley groups, with an eye towards the specific subgroups used to build Chevalley coset complex HDXs [24, 25, 34, 35]. Our definitions start out somewhat abstract; see Section 3.1 for a concrete example.

Chevalley groups [8, 10, 22, 39] are best known as one of the four classes of finite simple groups [2]. Simple groups have no nontrivial normal subgroups, and thus they have no nontrivial quotient groups. As a result, simple groups are to finite group theory what prime numbers are to number theory.

Chevalley groups are related to the more-famous Lie groups. A *Lie group* is a group on matrices which admits a certain good topological structure. For example, the special linear groups (i.e., groups of determinant-1 square matrices of a fixed size) over $\mathbb{R}$ and $\mathbb{C}$ are Lie groups. Roughly, Chevalley groups are Lie groups defined over finite fields, such as the special linear group over $\mathbb{F}_q$ (where $\mathbb{F}_q$ denotes the field with q elements for q a prime power).

One way of capturing how these groups behave is with a set of generators on a particular root system. As it turns out, every Chevalley (and Lie) group can be expressed in this way.

▶ **Definition 2** (Root system; generators). *Let G be a Chevalley (or Lie) group. Then G can be associated with a finite set of nonzero vectors $\Psi \subset \mathbb{R}^n$. Each vector $\zeta \in \Psi$ is called a* root*, and the set Ψ is called a* root system*. For each root $\zeta \in \Psi$ and element $t \in \mathbb{F}_q$ (or $\mathbb{R}$ or $\mathbb{C}$, for Lie groups), there exists a* generator *$\{\zeta, t\} \in G$.*

True to their name, the set of all such $\{\zeta, t\}$ generates G, and the algebraic structure of G with respect to these generators (i.e., what happens when you multiply two generators together) is intimately connected to the geometric structure of Ψ.

While the generators $\{\zeta, t\}$ have a deeper meaning (see Section 3.1), we can interpret them for now as abstract symbols satisfying a set of equations, called *Steinberg relations*, that describe how the generators behave under multiplication. There are three kinds of Steinberg relations, based on whether the roots of the generators are the same, distinct, or of opposite sign, with each case giving a different kind of relation. These three kinds are, respectively: (i) linearity relations, (ii) commutator relations, and (iii) diagonal relations.

The linearity relations are the simplest: when two generators share a root, multiplication commutes with field addition. In other words, for every $t, u \in \mathbb{F}_q$, $\{\zeta, t\} \cdot \{\zeta, u\} = \{\zeta, t + u\}$.

The commutator relations describe how generators with distinct roots multiply together by specifying their commutator (hence the name). In a group, the *commutator* of two elements g and h is defined as $[g, h] := ghg^{-1}h^{-1}$, or equivalently, $gh = [g, h]hg$. Intuitively, the commutator is a measure of how much g and h fail to commute. For example, g and h commute iff $[g, h] = \mathbb{I}$, where $\mathbb{I}$ is the identity element of the group.

As it turns out, if $\eta \neq \pm\zeta \in \Psi$ are two distinct roots, then every commutator of Chevalley generators $[\{\zeta, t\}, \{\eta, u\}]$ can be written as a product of generators of the form $\{a\zeta + b\eta, C^{a,b}_{\zeta,\eta} t^a u^b\}$, where $a, b \in \mathbb{N}^+$ and where $C^{a,b}_{\zeta,\eta} \in \{\pm 1, \pm 2, \pm 3\}$ is a particular *Chevalley constant* depending on ζ, η, a, and b.[2] Remarkably, the generators in these products are always contained in the positive integer span of the two original roots η and ζ. As a result, the commutator relations connect the algebraic structure of the Chevalley group to the geometry of its root system.

The diagonal relations cover the final case, where $\eta = -\zeta$. They are less convenient to state, but luckily for us, we can ignore them. This is because HDX constructions only use unipotent Chevalley subgroups, whose diagonal Steinberg relations are trivial.[3]

▶ **Definition 3** (Unipotent; positive root system). *A Chevalley subgroup is* unipotent *if its root system is positive. Let Ψ be a root system, and let $\zeta_1, \ldots, \zeta_k \in \Psi$ be a set of linearly-independent roots, which we call a* base. *(Note that $k < n$ is possible, i.e., $\zeta_1, \ldots, \zeta_k$ need not form a basis of $\mathbb{R}^n$ in the linear-algebraic sense.) Then the* positive root system Φ *is the set of vectors in Ψ that lie in the nonnegative span of the base vectors:*

$$\Phi := \left\{ \sum_{i=1}^{k} a_i \zeta_i \in \Psi \;\middle|\; a_1, \ldots, a_k \in \mathbb{N} \right\}.$$

Given a positive root system Φ, there is a corresponding unipotent subgroup G of the Chevalley group that is generated by elements $\{\zeta, t\}$ for $\zeta \in \Phi$ and $t \in \mathbb{F}_q$. To define this subgroup with Steinberg relations, all that is needed are (i) linearity relations for every root in Φ, and (ii) commutator relations for every pair of distinct, non-opposite roots in Φ.

In addition to being unipotent, the Chevalley subgroups used to build HDXs are graded, and this gradedness is taken with respect to the heights of the roots.

▶ **Definition 4** (Heights). *Let Φ be a positive root system with base $\zeta_1, \ldots, \zeta_k$. We define a function* height $: \Phi \to \mathbb{N}^+$ *via* $\text{height}(\sum_{i=1}^{k} a_i \zeta_i) = \sum_{i=1}^{k} a_i$. *Note that, in particular,* $\text{height}(\zeta_i) = 1$ *for each base root ζ_i.*

[2] These constants (and their signs) have a reputation of making calculations on Chevalley groups inconvenient. The calculations in the KOOS proofs (and thus our formalization) were no exception.

[3] We only need to consider these subgroups due to the so-called *local-to-global* theorems for HDXs [17].

▶ **Definition 5** (Graded). *Let G be a Chevalley group associated with a root system Ψ over the finite field $\mathbb{F}_q = \mathbb{F}_p[X]/\mu$, where $q = p^m$ for prime power p and $m \in \mathbb{N}$, and where μ is an irreducible polynomial of degree $m+1$ in X. Then the generators of G can be written as $\{\zeta, f\}$, where $\zeta \in \Psi$ and f is a polynomial of degree at most m over $\mathbb{F}_q$. A* graded unipotent *subgroup of G is spanned by the generators $\{\zeta, f\}$, where f has degree at most* height(ζ).

An important feature of this definition is that, given a fixed positive root system, the graded unipotent subgroup does not depend on m, so long as m is larger than the maximum height of any root. This parameter m does not appear in our formalization, but when thinking about HDX constructions, m guarantees that we can make an infinite family of increasingly large groups, which are used to make an infinite family of increasingly large simplicial complexes (i.e., HDXs).

We simplify this setup slightly and restrict our graded generators, without loss of generality, to just those where f is a monomial. We write these generators as triples $\{\zeta, i, t\} := \{\zeta, tX^i\}$, where $0 \le i \le m$ is the *degree* of the monomial and $t \in \mathbb{F}_q$ is its *coefficient*. Using this notation, the graded unipotent Steinberg relations are now:

- Linearity: For every $\zeta \in \Phi$, $0 \le i \le m$, and $t, u \in \mathbb{F}_q$, $\{\zeta, i, t\} \cdot \{\zeta, i, u\} = \{\zeta, i, t+u\}$. (A direct consequence of the linarity relations are the *identity* relations, where $\{\zeta, i, 0\} = \mathbb{I}$; and the *inverse* relations, where $\{\zeta, i, t\}^{-1} = \{\zeta, i, -t\}$.)
- Commutator: For every $\zeta \neq \eta \in \Phi$, $0 \le i, j \le m$, and $t, u \in \mathbb{F}_q$, the commutator of $\{\zeta, i, t\}$ and $\{\eta, j, u\}$ is a product of generators of the form $\{a\zeta + b\eta, ai + bj, C_{\zeta,\eta}^{a,b} t^a u^b\}$.
- *Mixed-degree*: For every $\zeta \in \Phi$, $0 \le i, j \le m$, and $t, u \in \mathbb{F}_q$, $[\{\zeta, i, t\}, \{\zeta, j, u\}] = \mathbb{I}$. (These relations are a consequence of the monomial restriction, and they encode the fact that addition of monomials is commutative.)

Given all this, we can finally define the unipotent subgroups of the graded Chevalley groups that we used in our formalization. The unipotent subgroup we want is the subgroup G generated by the elements $\{\zeta, i, t\}$, where $\zeta \in \Phi$, $t \in \mathbb{F}_q$, and $0 \le i \le$ height(ζ). From a Steinberg-relational perspective, G is defined by the linearity, commutator, and mixed-degree relations involving roots in Φ.

The proof of Theorem 1, then, is to show that this particular set R of linearity, commutator, and mixed-degree Steinberg relations (and a few more; see Section 3.3) implies the rest. This amounts to showing that any relation not in R can be derived using relations in R, along with general facts about groups and the underlying field $\mathbb{F}_q$.

The technical details for why Theorem 1 is necessary in order to build HDXs are beyond the scope of this paper. But extremely roughly, when building an HDX from one of these Chevalley groups, one builds a coset complex CC with a graded Chevalley group G and certain designated subgroups of G, where the subgroups are defined by the relations in R. Showing that CC is an HDX is equivalent to showing that every Steinberg relation for G has a short derivation using the relations in R. That's where the statement "R implies the rest of the Steinberg relations" comes in.

3.1 A concrete example: B_n and B_3

Let's now see these definitions in action. We consider the B_n-type Chevalley groups in general and the B_3-large case in particular. Since these groups are defined over matrices, we will need the following matrix notation. Let $e_1, \ldots, e_n$ be the standard basis for $\mathbb{R}^n$, let $\mathbb{I}_n$ be the $n \times n$ identity matrix, and let $E_{i,j}$ be the *indicator matrix* with a 1 in the (i, j) entry and 0s everywhere else. In all cases, the group operation $(\cdot)$ is matrix multiplication.

The B_n-type groups have as elements (some of the) $(2n+1)\times(2n+1)$ orthogonal matrices with determinant 1. To emphasize the symmetry of the matrix entries about the middle row and column, we label the rows and columns from top to bottom and from left to right with an index between $-n$ and n, which we write as $[\pm n]$. Traditionally ([6, §11.3], following [11,36]), the matrix entries are elements of a finite field $\mathbb{F}_q$, but as we have already discussed, the graded versions of the entries are degree-bounded polynomials in a single variable over $\mathbb{F}_q$. For the purposes of this example, we will stick with the entries being elements of $\mathbb{F}_q$.

In order to state the Steinberg relations for the unipotent B_n groups, we need a positive root system Φ. We start with the eponymous B_n set, which serves as our (not positive) root system Ψ. The set $B_n \subset \mathbb{R}^n$ comprises all *short* and *long* roots, which are vectors supported by one and two ± 1 coordinates, respectively. In other words, the roots can be written as ae_i and $ae_i + be_j$, respectively, for $a, b \in \{\pm 1\}$ and $1 \le i\ (< j) \le n$.

Given this root system, we identify our generators $\{\zeta, t\}$ with the following matrices. It is these matrices that actually generate the group.[4]

$$\{ae_i, t\} \coloneqq \mathbb{I}_{[\pm n]} + 2atE_{ai,0} - atE_{0,-ai} - t^2E_{ai,-ai}, \qquad \text{(short root)}$$
$$\{ae_i + be_j, t\} \coloneqq \mathbb{I}_{[\pm n]} + atE_{ai,-bj} - atE_{bj,-ai}. \qquad \text{(long root)}$$

In the B_3 case, the group elements are 7×7 matrices. Below are two examples of generators (with a matrix entry of $\cdot$ meaning 0).

$$\{e_1 - e_2, t\} = \begin{pmatrix} 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & -t & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & +t & \cdot & \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \end{pmatrix}, \quad \{e_2, t\} = \begin{pmatrix} 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \\ \cdot & -t & \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot \\ \cdot & -t^2 & \cdot & +2t & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \end{pmatrix}.$$

Now that we have the set of generators for the root system $\Psi =: B_n$, we can restrict this set to those for a positive root system Φ. Doing so allows us to ignore the diagonal Steinberg relations, the effect of which makes the subgroup simpler and easier to work with. To make such a restriction, we must pick a base of linearly-independent roots from Ψ.

In the B_3 case, we care about four particular roots (two long, two short):

$$\alpha := e_1 - e_2, \qquad \beta := e_2 - e_3, \qquad \psi := e_3, \qquad \omega := -e_1.$$

The seven roots in the B_3-small positive root system are spanned by the base $\{\beta, \psi, \omega\}$, while the nine roots in the B_3-large positive root system are spanned by the base $\{\alpha, \beta, \psi\}$. (Hence the names "small" and "large.") For completeness, we list the nine B_3-large roots here:

$$B_3\text{-large} := \{\alpha,\ \beta,\ \psi,\ \alpha+\beta,\ \beta+\psi,\ \beta+2\psi,\ \alpha+\beta+\psi,\ \alpha+\beta+2\psi,\ \alpha+2\beta+2\psi\}.$$

The Steinberg relations each involve a commutator on a pair of $\{\zeta, t\}$ generators. Earlier, we claimed that in a unipotent Chevalley group, each commutator of generators can be written as a product of generators involving Chevalley constants and positive integer spans.

[4] The astute reader may notice that these matrices are not orthogonal in the standard sense, i.e., they may not preserve the squared distance $\sum_{i=-n}^{n} v_i^2$ of a vector $v \in F^{[\pm n]}$. However, they do preserve a different quadratic form, namely, $\sum_{i=-n}^{n} v_i v_{-i}$. And since all quadratic forms are equivalent when F is a field of odd characteristic (which we require), then we can isomorphically map these matrices into ones which preserve the usual squared distance, though these do not have as succinct a form.

For example, consider $[\{\alpha, t\}, \{\beta + \psi, u\}]$ (which are the $\{e_1 - e_2, t\}$ and $\{e_2, t\}$ matrices above). One can verify by hand that

$$
\begin{aligned}
&[\{\alpha, t\}, \{\beta + \psi, u\}] = [\{e_1 - e_2, t\}, \{e_2, u\}] \\
&= \begin{pmatrix}
1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & +tu & 1 & \cdot & \cdot & \cdot \\
\cdot & \cdot & -(tu)^2 & -2tu & 1 & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1
\end{pmatrix}
\cdot
\begin{pmatrix}
1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & 1 & +tu^2 & \cdot \\
\cdot & \cdot & \cdot & \cdot & -tu^2 & 1 & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1
\end{pmatrix} \\
&= \{\alpha + \beta + \psi, tu\} \cdot \{\alpha + 2\beta + 2\psi, tu^2\}.
\end{aligned}
$$

Indeed, the commutator of these two generators is a product of generators in their roots' positive span! This is precisely the commutator relation for this pair of generators.

From this example, one can hopefully appreciate the value of representing the underlying group using Steinberg relations. Expressed in matrices, the equation requires expanding several matrix multiplications, i.e., $[\{\alpha, t\}, \{\beta+\psi, u\}] = \{\alpha, t\}\{\beta+\psi, u\}\{\alpha, t\}^{-1}\{\beta+\psi, u\}^{-1}$. But on the level of generators, the equation just expresses a simple "combinatorial" fact about the span of some vectors in a particular finite set. That the entire structure of matrix groups may be captured by such simple relations is a kind of mathematical miracle (and is the basis for Lie group theory).

As we discussed just before this subsection, our formalization involves showing that a subset of the Steinberg relations implies the rest. It is no exaggeration to say that almost all of our lemmas take a handful of relations and rewrite them into the form of a "missing" relation. To give a taste of what these lemmas look like, we close this section by considering the linearity relations in the B_3-small case for the generators $\{\zeta, i, t\}$ with root $\zeta := \beta+\psi+\omega$.

Our goal is to show that the linearity relations for ζ hold; that is, for all $0 \leq i \leq m$ and for all $t, u \in \mathbb{F}_q$, $\{\zeta, i, t\}\{\zeta, i, u\} = \{\zeta, i, t + u\}$. (In this example, we omit the group operation symbol $(\cdot)$.) As we discuss in Section 3.3, in addition to the linearity, commutator, and mixed-degree relations on the roots in Φ, we may assume the following relations:

1. For all $i \leq 3$, $j \leq 1$, and $t, u \in \mathbb{F}_q$, $[\{\beta + \psi + \omega, i, t\}, \{\beta, j, u\}] = \mathbb{I}$. (I.e., they commute.)
2. For all $i \leq 3$, $j \leq 2$, and $t, u \in \mathbb{F}_q$, $[\{\beta + \psi + \omega, i, t\}, \{\psi + \omega, j, u\}] = \mathbb{I}$.
3. For all $i \leq 1$, $j \leq 2$, and $t, u \in \mathbb{F}_q$, $\{\beta + \psi + \omega, i + j, tu\} = [\{\beta, i, t\}, \{\psi + \omega, j, u\}]$. Equivalently, $\{\beta + \psi + \omega, i + j, tu\} = \{\beta, i, t\}\{\psi + \omega, j, u\}\{\beta, i, -t\}\{\psi + \omega, j, u\}$, via $[g, h] = ghg^{-1}h^{-1}$ and the inverse relations for the smaller roots.

Decompose $i = i_1 + i_2$ arbitrarily into $0 \leq i_1 \leq 2$ and $0 \leq i_2 \leq 1$.

$$
\begin{aligned}
&\{\beta + \psi + \omega, i, t\}\{\beta + \psi + \omega, i, u\} \\
&= \{\beta, i_i, t\}\{\psi + \omega, i_2, 1\}\{\beta, i_1, -t\}\{\psi + \omega, i_2, -1\}\{\beta + \psi + \omega, i, u\} \\
&\qquad \text{(Equivalent def. in (3), } tu := t \cdot 1) \\
&= \{\beta, i_1, t\}\{\beta + \psi + \omega, i, u\}\{\psi + \omega, i_2, 1\}\{\beta, i_1, -t\}\{\psi + \omega, i_2, -1\} \\
&\qquad \text{(Commute } \{\beta + \psi + \omega, j, u\} \text{ to the left via items (1) and (2))} \\
&= \{\beta, i_1, t + u\}\{\psi + \omega, i_2, 1\}\{\beta, i_1, -u\}\{\psi + \omega, i_2, -1 + 1\}\{\beta, i_1, -t\}\{\psi + \omega, i_2, -1\} \\
&\qquad \text{(Equivalent def. in (3), } tu := u \cdot 1\text{; absorb terms 1 and 4 via linearity)} \\
&= \{\beta, i_1, t + u\}\{\psi + \omega, i_2, 1\}\{\beta, i_1, -(t + u)\}\{\psi + \omega, i_2, -1\} \\
&\qquad \text{(Identity relation on } \psi + \omega\text{; linearity relation on } \beta) \\
&= \{\beta + \psi + \omega, i, t + u\}. \qquad \text{(Equivalent def. in (3), } tu := (t + u) \cdot 1)
\end{aligned}
$$

3.2 Presentations of groups

So far, we have been informal when we say that (graded) (unipotent) Chevalley groups can "be defined" by the Steinberg relations. What we mean to say is that the relations and the generators form a presentation of these groups.

In abstract algebra, a *presentation* is a way to specify a group with a set of relations. Let S be a finite set of symbols, called *generators*; let S^{-1} be their inverses; let $\mathbb{S} := S \cup S^{-1}$ be our alphabet; and let $R \subseteq \mathbb{S}^*$ be a set of finite-length strings, called *relations*. Then a group $G = \langle S \mid R \rangle$ can be specified as follows:

- The elements of G are equivalence classes of strings in $\mathbb{S}^*$, where $w \sim w'$ iff w can be transformed into w' via a finite sequence of additions and deletions of the identity-equivalent substrings ss^{-1}, $s^{-1}s$, and $r \in R$ anywhere in the string.
- The group operation $(\cdot)$ of G is string concatenation.
- The identity element $\mathbb{I}$ is the empty string ϵ.
- The inverse $(s_1 \cdots s_k)^{-1}$ is $s_k^{-1} \cdots s_1^{-1}$, where $s_1, \ldots, s_k \in \mathbb{S}$ and $(s^{-1})^{-1} = s$.

This definition is commonly used in combinatorial group theory. An equivalent definition (used in `mathlib`) is that G is the free group on S modulo the normal closure of R in S.

The simplest nontrivial presentation is where $S = \{x\}$ and $R = \varnothing$. In this case, every string in $\mathbb{S}^*$ can be simplified to $x^{\pm k}$ by deleting any xx^{-1} and $x^{-1}x$ substrings. For example, $xxx^{-1}xxxx^{-1}x^{-1}$ simplifies to xx. These simplified strings are isomorphic to the integers, with concatenation isomorphic to addition, and thus we write $\langle S \mid R \rangle \cong \mathbb{Z}$.

Some variations on this example are:

- $\langle \{x\} \mid \{x^n\} \rangle \cong \mathbb{Z}_n$, the integers modulo n under addition.
- $\langle \{x, y\} \mid \{xyx^{-1}y^{-1}\} \rangle \cong \mathbb{Z} \times \mathbb{Z}$, the group of integer pairs under addition. To see why, notice that any yx substrings can be replaced with xy via a single left-multiplication of $xyx^{-1}y^{-1}$, and so the canonical representation of these strings is $x^{\pm k}y^{\pm m}$. (Note that $xyx^{-1}y^{-1} = [x, y]$. By declaring the commutator of x and y as a relation, we are ensuring that x and y commute in the resulting group.)

Presentations often give a succinct way of specifying a group when its multiplication table is too large to write down. In our case, given a field $\mathbb{F}_q$ and a root system Φ, an (ungraded) Chevalley group has $q^{|\Phi|}$ elements, which means that its multiplication table has $q^{|\Phi|^2}$ entries. In contrast, there are only about $|\Phi|^2 q^2$ Steinberg relations (or rather, $|\Phi|^2$ relations, each parameterized by any pair of the q field elements), which is much more manageable.

Using the language of presentations, KOOS (and we) were interested in the following problem: given a root system Φ and its Steinberg relations R, determine if there is a smaller set of relations R' that presents an isomorphic group, i.e., where $\langle S \mid R \rangle \cong \langle S \mid R' \rangle$. Since groups presented by fewer relations tend to be larger, due to fewer elements being "modded out," this is equivalent to showing that R can be derived from R'.

3.3 Our theorem, restated

Now that we have defined graded unipotent Chevalley groups and introduced the notion of group presentations, we can formally restate our main theorem. This theorem analyzes presentantions of three graded unipotent Chevalley groups: the A_3 group, the B_3-small group, and the B_3-large group. (The former was analyzed by KO [25] and the latter two by OS [35].) Each of these groups is associated with three base roots and a corresponding root system. Specifically, each group is defined by a presentation, which we call the *full* presentation, which includes (i) linearity relations for every root, (ii) commutator relations

for every pair of distinct roots, and (iii) mixed-degree relations for every root. We write this presentation as $\langle S \mid R_{\text{full}} \rangle$. These presentations and groups are defined concretely in our Lean code as `fullA3Graded`, `fullB3SmallGraded`, and `fullB3LargeGraded`.

The theorem we prove shows that, in all three cases, not all Steinberg relations are necessary to define the group. In other words, one can give a presentation of these groups using a smaller set of relations R_{weak}, which we call the *weak* set. These "weak" relations fall into three types:

1. The *codimension-1* relations. A root is *present* if it is in the span of any two of the three base roots, and *missing* otherwise. For example, in the B_3-large case described in Section 3.1, $\alpha+\beta$ is present, but $\alpha+\beta+2\psi$ is missing. The *codimension-1 relations* include the linearity and mixed-degree relations for every present root, as well as the commutator relations for every pair of distinct present roots. For example, the commutator relation for the pair $(\alpha, \alpha+\beta)$ is included (since these are in the nonnegative span of the base roots α and β), but the commutator relation for $(\alpha+\beta, \beta+\psi)$ is not.
2. The *lifted* relations. Recall the definition of the ungraded Chevalley group in Section 3. Letting G' be the corresponding ungraded unipotent subgroup (i.e., A_3, B_3-small, B_3-large resp.), one can build so-called lifting homomorphisms $G \to G'$ from the ungraded group to the graded group given by mapping each generator to a generator of the same root type. The *lifted* relations are images in G of relations involving the present roots (not necessarily Steinberg relations) in G' under these lifting homomorphisms.
3. The *definition* relations. The first two types of relations only involve generators with present roots. In order to derive relations involving generators with missing roots $\{\zeta, i, t\}$, we add a relation $\{\zeta, i, t\}^{-1}e$ for some expression e on present-root generators such that the relation "$\{\zeta, i, t\} = e$" is derivable from the full set of Steinberg relations. (There are many possible choices for e, but we fix one specific choice in our formalization for convenience. We leave the proof that e can be arbitrary for future work.)

These presentations and groups are defined concretely in our Lean code as `weakA3Graded`, `weakB3SmallGraded`, and `weakB3LargeGraded`.

With all of this background in hand, we can now restate our main theorem.

▶ **Theorem 1** (Revisited; [25, 35]). *Let G be one of three specific graded unipotent Chevalley groups (A_3, B_3-small, or B_3-large), and let Φ be a positive root system for G. Then the full set of Steinberg relations R_{full} for G can be derived from the weak relations R_{weak}. In other words, the presentations $\langle S \mid R_{full} \rangle$ and $\langle S \mid R_{weak} \rangle$ are isomorphic.*

4 Our formalization: definitions and top-level theorem statement

In this section, we present the Lean versions of our definitions and top-level theorems. We also discuss some design decisions we made and how they affected our verification.

Our main result, Theorem 1, is stated and proven purely using the language and methods of group theory. And since `mathlib` has good support for abstract algebra, our formalization was relatively straightforward to set up, with our definitions of root systems, generators, and presentations mapping onto key definitions already present in `mathlib`. Of course, we still had to muck our way through the hundreds of lemmas from the KOOS papers, but it was the proofs, not the setup, where we spent most of our time.

First, we define our three root systems. Recall that our three graded unipotent Chevalley groups are each associated with a positive root system Φ imbued with a `height` function. In Lean, we use a type class so that we can state things about root systems in general, and then we show that each specific system is an instance of this class. For example, here are the definitions for the B_3-large case.

```
class PositiveRootSystem (Root : Type) where
  height : Root → Nat

inductive B3LargePosRoot
  | α | β | ψ | αβ | βψ | β2ψ | αβψ | αβ2ψ | α2β2ψ

def B3Large.height : B3LargePosRoot → Nat
  | α | β | ψ => 1
  | αβ | βψ => 2
  | ...   /- For brevity, we omit the other cases -/

instance : PositiveRootSystem B3LargePosRoot where height := B3Large.height
```

On paper, we would write the linear combinations of roots with "+." For example, $\alpha\beta$ is typically written as "$\alpha + \beta$." But Lean makes it hard to use reserved symbols in definition names, so we drop the plus signs in the names of the roots in `B3LargePosRoot`.

Once we have a particular root system and a ring/field for the coefficients, we can define the type[5] of the $\{\zeta, i, t\}$ triples, which serve as the generators of the (graded) presentation.

```
structure GradedChevGen (Φ R : Type) [PosRootSys Φ] [Ring R] where
  ζ : Φ
  i : ℕ
  hi :  i ≤ height ζ
  t : R
```

An analogous type `ChevalleyGenerator` represents the ungraded generators $\{\zeta, t\}$ by omitting the `i` and `hi` fields.

However, we want to view the elements of `GradedChevGen` as elements of a free group, and not as a standalone type. Thus, we need to inject them into the free group on T. Fortunately, the `FreeGroup` type in `mathlib` already has the needed definitions and theorems. In this case, the function `FreeGroup.of : T → FreeGroup T` performs the injecting.

Next, we define the presentations for each Chevalley group from their Steinberg relations. To do so, we use `mathlib`'s definition of a `PresentedGroup`, which is the free group on a type T, quotiented by a set of relations written in terms of the free group on T.

```
def PresentedGroup (rels : Set (FreeGroup T)) :=
  FreeGroup T / Subgroup.normalClosure rels
```

Armed with our definitions of the weak and full sets of relations (which we will discuss shortly), we can state our top-level theorems. For example, here is the theorem statement for the B_3-large case:

```
theorem full_relations_implied_by_weak_relations :
  ∀ r ∈ fullB3LargeGraded.allRelations, weakB3LargeGraded.project r = 1 := ...
```

In other words, all relations in the full set vanish in the group presented by the weak set. (`project` projects free group elements into the presented graded Chevalley group.)

[5] In mathematics, groups and other objects are defined over sets, but in Lean, the same objects are defined over *types*. To be more consistent with Lean's view of things, we will refer to "types" over "sets." We apologize to the mathematicians.

As is the case for many formalizations, this succinct theorem statement hides much of the complexity of the underlying definitions. We leave the details for those who look at our Lean files, but we highlight three features that are characteristic of working with Chevalley groups in particular and `mathlib`'s version of presented groups in general.

The first feature is that we decided to manually define the additive structure of the positive root systems Φ. Ideally, these systems would be implicitly defined by their base, but to define "+" in Lean, we would need to somehow encode the fact that addition on roots is not total, since the sum of any two roots is not necessarily a new root. For example, in the B_3 case, $\alpha + 3\beta$ is a positive linear combination of α and β, but it is not a root because $\alpha + 3\beta \notin B_3$ (i.e., it is not a long or a short root). So instead of carrying around hypotheses that any particular addition is sound, or defining `add : Φ → Φ → Option Φ` and enforcing that the result be `some ζ`, we write down all of the needed roots and ensure that their heights are well-behaved by construction.

In our opinion, our decision to manually list the roots made our formalization easier. Because we had to use one of the listed roots, Lean's type checker was quick to point out when we wrote down the wrong root or the wrong Chevalley constant, and it was easy to prove that the degrees i of graded generators $\{\zeta, i, t\}$ were less than the height of ζ. (We wrote a custom tactic `ht` that unfolded `height` and then called the `omega` tactic, which is a tactic for solving systems of linear inequations on integers.) That being said, we recognize that this choice becomes less beneficial as the size of the root system increases. If our work continues, then this choice is a good candidate for a refactor.

The second feature we highlight is that our definitions for the various (graded, unipotent) Chevalley groups overlap in a complicated way. We made this design choice in an attempt to reduce code duplication and to abstract out common properties shared by the different groups. While this makes our code harder to understand at first glance, we hope that it provides a good basis for future Chevalley formalizations.

This overlap is most obvious in our definitions of the weak and full sets of relations. Recall that the full set of unipotent Steinberg relations comprises the linearity and commutator relations, while the weak set comprises codimension-1, lifted, and definition relations. To capture the set of Steinberg relations shared by both sets, we define the following:

```
structure PartialChevalleySystem (Φ : Type) [PosRootSys Φ] where
  presentRoots : Set Φ
  trivialSpanRootPairs : Set (Φ × Φ)
  singleSpanRootPairs : Set (SingleSpanRootPair Φ)
  doubleCommutatorRootPairs : Set (DoubleSpanRootPair Φ)
  ...  /- hypotheses ensuring that the relations only involve present roots -/
```

These four fields contain the relations shared between the two sets, namely, the linearity and mixed-degree relations for the present roots and the commutator relations for pairs of roots spanning zero, one, and two additional roots in the root system.[6] (The types `Single-` and `DoubleSpanRootPair` encode these additional root(s) and their Chevalley constants. We supply them by hand, and then later we verify that they are correct.) The set `presentRoots` contains every root in the full case, but only the present roots in the weak case.

In the weak case, the `PartialChevalleySystem` lacks the lifted and definition relations. We add them separately in a new type `GradedPartialChevalleyGroup`. The lifted relations, we supply directly; the definition relations, we specify via a function `define`. We require that

[6] The division of the relations into these four types is specialized for the A_n and B_n systems. More-complicated systems would need more fields.

`define` acts as the identity function on all of the present roots, and that it commutes with `FreeGroup.lift` (which extends functions `T → [Group G]` to a group homomorphism from `T` to `G`). We bundle these two additional kinds of relations as follows:

```
structure GradedPartialChevalleyGroup (Φ R : Type) [PosRootSys Φ] [Ring R] where
  sys : PartialChevalleySystem Φ
  liftedRels : Set (FreeGroup (GradedChevGen Φ R))
  define : GradedChevGen Φ R → FreeGroup (GradedChevGen Φ R)
  ...  /- hypotheses ensuring that 'define' is well-behaved -/
```

Recall that the definition relations need an expression e. It is in `define` that we choose this e for each missing root. For example, `define` for the weak B_3-large case looks like this:

```
def weakB3Large.define (F : Type) [Field F] (g : GradedChevGen B3LargePosRoot F)
    : FreeGroup (GradedChevalleyGenerator B3LargePosRoot F) :=
  let ⟨ ζ, i, hi, t ⟩ := g;
  match ζ with
  | αβψ    => {βψ, (split3 i).2, -1/2} * {α, (split3 i).1, t}
              * {βψ, (split3 i).2, 1} * {α, (split3 i).1, -t}
              * {βψ, (split3 i).2, -1/2}
  | αβ2ψ   => [ {α, (split4 i).1, t},   {β2ψ, (split4 i).2, 1} ]
  | α2β2ψ  => [ {αβ, (split5 i).1, -t}, {β2ψ, (split5 i).2, 1} ]
  | _ => FreeGroup.of g
```

(The `splitX` functions send $i \leq \text{height}(\zeta)$ to a pair (j, k), where $j + k = i$ and where j and k are less than the other relevant root heights. Basically, `splitX` acts as an explicit partitioning function to factor the X^i monomials into two monomials for two different generators.)

In the full case, these two additional fields are trivial: `liftedRels` is the empty set, and `define` is the identity function.

The third feature we want to highlight is how the definition of a presentation in Lean often makes it awkward to use relations when reasoning about the presented group. There are three reasons for why this is the case.

The first reason is that all of the relations need to be in one set, but we usually want to focus on a specific subset. When defining a `PresentedGroup`, we have to lump the relations into one large `Set`. But as we have already seen, we define different subsets of relations according to their shared properties (e.g., the trivial span relations), and in order to talk about these shared properties (e.g., to assert that one *specific* pair of generators satisfies a trivial commutator relation), we have to "unpack" this large set of relations into its component pieces. To solve this problem, we wrote a lemma to do this for each kind of subset.

The second reason for this awkwardness is that relations are written in terms of the free group, but we often want to work with equations inside the presented group. These are equivalent notions, since a relation r can be thought of as $r \sim \mathbb{I}$ in the presentation or as $r = \mathbb{I}$ in the presented group. But while these are equivalent, it is usually easier to work with the equation view, especially for our purposes, where we are trying to show that some Steinberg relations can be derived from others. In the equation view, this amounts to showing that a relation not in the weak set is equal to $\mathbb{I}$ in the presented group, which then allows us to use all of the typical lemmas and machinery associated with group theory.

The third awkward reason is that we often have to convert the relations into a form that is easier to use in proofs. The harder-to-use form is a consequence of how presentations are defined. In a presentation, every relation r encodes $r = \mathbb{I}$, (or rather, $r \sim \mathbb{I}$). But we often want to view a relation as a pair of elements (r_1, r_2) such that $r_1 = r_2$ (via $r_1 r_2^{-1} =: r = \mathbb{I}$).

This allows us to state the commutator relations more naturally and to use them in Lean proof tactics, such as `rw` and `simp`. For example, the commutator relations for a pair of roots spanning two additional roots (the `DoubleSpanRootPair`) is usually written as

$$[\{\zeta, i, t\}, \{\eta, j, u\}] = \{\theta_1, i+j, C^{1,1}_{\zeta,\eta} tu\} \cdot \{\theta_2, i+2j, C^{1,2}_{\zeta,\eta} tu^2\},$$

but to express it in the $r = 1$ form, the right-hand side of the expression needs to be inverted and moved over to the left. As a result, our Lean definition is:

```
def doubleSpanRelationsOfRootPair (R : Type) [Ring R] (p : DoubleSpanRootPair Φ)
    : Set (FreeGroup (GradedChevGen Φ R)) :=
  let ⟨ ζ, η, θ₁, θ₂, C₁, C₂ ⟩ := p;
  { [ {ζ, i, t}, {η, j, u} ]
      * ({θ₁, i + j, C₁ * t * u} * {θ₂, i + 2 * j, C₂ * t * u * u})⁻¹
    | (i j : ℕ) (i ≤ height ζ) (j ≤ height η) (t u : R) }
```

Clearly, this isn't as nice. So we also have an equivalent definition that restates these relations in the $r_1 = r_2$ form, as well as a lemma that converts between the two.

5 Our formalization: the shape of the proof

In the introduction, we wrote that the KOOS proofs (and our formalization) involved a lot of casework, and unfortunately, we do not know of a way to simplify their proof approach. Simply put, to show that the full set of Steinberg relations is derivable from the weak set, we showed one-by-one that every missing Steinberg relation holds in the group presented by the weak relations. To put some numbers to it, of the commutator relations (which are more numerous and harder to show than the linearity and the mixed-degree relations), 15 are missing in A_3, 21 are missing in B_3-small, and 26 are missing in B_3-large.

Roughly, most of the missing relations involve one or more missing roots. In the A_3 and B_3-small cases, there is only one missing root, but in the B_3-large case, there are three missing roots: $\alpha+\beta+\psi$, $\alpha+\beta+2\psi$, and $\alpha+2\beta+2\psi$. The key to the proofs is to express the generator involving the missing root in terms of generators of present roots, but there may be several ways of doing so. For example, there are three ways to express $\alpha+2\beta+2\psi$ as a sum of two present roots:

$$\alpha+2\beta+2\psi = (\alpha+\beta)+(\beta+2\psi) = (\alpha+\beta+2\psi)+\beta = (\alpha+\beta+\psi)+(\beta+\psi).$$

Ultimately, we show that any of these forms works. Then, we use the relations involving the present roots we already have to adjust the group expressions until both sides are equal. This is where the term rewriting problems come from.

To express the missing root ζ in terms of smaller roots, we apply the definition relation for ζ, which states that $\{\zeta, i, t\} = e$ for some fixed expression e on present-root generators. But because we should not depend on the choice of e (too much), we show that a set of *interchange* theorems holds, which allows us to shuffle the degrees and coefficients of the generators to the form most convenient for the relation we are trying to prove.

Here's an example of an interchange theorem:

```
theorem interchange_αβψ_βψ_α :  ∀ (i j k : ℕ) (t u : F),
    ⟪βψ, j+k, -1/2⟫ * ⟪α, i, t*u⟫ * ⟪βψ, j+k, 1⟫ * ⟪α, i, -t*u⟫ * ⟪βψ, j+k, -1/2⟫
  = ⟪βψ, j+k, -u/2⟫ * ⟪α, i, t⟫ * ⟪βψ, j+k, u⟫ * ⟪α, i, -t⟫ * ⟪βψ, j+k, -u/2⟫
```

Informally, this is almost expressing the commutator $[\{\alpha, i, tu\}, \{\beta+\psi, j+k, 1\}]$, but with the inverses pushed into the generators (via the inverse relations $\{\zeta, i, t\}^{-1} = \{\zeta, i, -t\}$, which are implied by the linearity relations for ζ) and with the coefficient of the rightmost inverse term ($\{\beta+\psi, j+k, -1\}$) split in half and added to the front and back of the expression. The precise reasons for why this theorem is true are not important; we merely provide it to show the nature of the interchange theorems, which all shuffle degrees and coefficients in this way.

Of the interchange theorems we need, we get eight of them almost for free from the lifting relations. This is due to how lifting works: when we map ungraded generators into the graded group, we get to pick a degree i for each of the three base roots ζ such that i is at most the height of ζ. In each of the groups we consider, the base roots all have height 1, meaning that $i \in \{0, 1\}$, and thus we have $2^3 = 8$ possible choices for the degrees.

As it turns out, each of the eight of these lifting relations look very similar to one of the interchange theorems, and so we only need to adjust the relations slightly to prove the theorems. We prove all of the other interchange theorems "the hard way" by solving a group rewriting problem.

Now armed with the definition relations and the interchange theorems, we can start proving that the rest of the missing Steinberg relations hold. We prove a majority of the relations "the hard way." For example, the proof from the end of Section 3.1, i.e., that the linearity relations hold for the missing root $\beta+\psi+\omega$, looks like this in Lean (albeit cleaned up and with several technical details omitted).

```
theorem lin_of_βψω : ∀ (i : ℕ) (t u : F),
    ⁅βψω, i, t⁆ * ⁅βψω, i, u⁆ = ⁅βψω, i, t + u⁆ := by
  intro i t u
  rcases decompose i with ⟨i₁, i₂⟩                    -- i = i₁ + i₂
  grw [ expr_βψω_as_β_ψω_β_ψω,                        -- Line 1
        expr_βψω_ψω_as_ψω_βψω, expr_βψω_β_as_β_βψω,
        expr_βψω_ψω_as_ψω_βψω,                         -- Line 2
        mul_one u, expr_βψω_as_β_ψω_β_ψω,             -- Line 3
        mul_one (t + u), expr_βψω_as_β_ψω_β_ψω ]      -- Lines 4 and 5
```

Notice how, despite the automation we used (see Section 6 for `grw`), the proof has the same form as the one we wrote by hand, with each step explicitly provided. (The `mul_one` theorem states that $x * 1 = x$, and is used to express $tu \coloneqq t \cdot 1$ and $tu \coloneqq (t+u) \cdot 1$.)

Unfortunately, to the best of our knowledge, many relations cannot be proven in this way. Rather, we have to prove that the relation holds for each possible concrete value for the pair of degrees i and j. For example, in the B_3-large case, we prove the following theorems in order to show that α and $\alpha + 2\beta + 2\psi$ commute:

```
theorem comm_of_α_α2β2ψ_00 : ∀ t u, [⁅α, 0, t⁆, ⁅α2β2ψ, 0, u⁆] = 1 :=
theorem comm_of_α_α2β2ψ_01 : ∀ t u, [⁅α, 0, t⁆, ⁅α2β2ψ, 1, u⁆] = 1 :=
theorem comm_of_α_α2β2ψ_02 : ∀ t u, [⁅α, 0, t⁆, ⁅α2β2ψ, 2, u⁆] = 1 :=
...  /- And nine more of these -/
```

(As we said, there's a lot of casework.)

The good news is that the graded unipotent Chevalley groups have a property called *reflection symmetry*. The Steinberg relations are closed under replacing every element $\{\zeta, i, t\}$ with the element $\{\zeta, \text{height}(\zeta) - i, t\}$. In other words, the degree can be "reflected" about the height of ζ. OS [35] (and we) use this property to derive half of these cases from the other half, which reduces the casework for these relations by a factor of two.

6 Our formalization: macros and other automation

Given the repetitive nature of the theorems and proofs in our formalization, it should come as no surprise that we developed custom macros, tactics, and proof automation to assist us. This automation falls into three categories: custom `simp` attributes, theorem macros, and group reassociation. None of these are particularly original, but they were quite effective in speeding up the pace of our development.

In Lean, invoking the simplifier tactic `simp` causes it to apply a library of lemmas and theorems marked with the `@[simp]` attribute in order to reduce the proof term to a "`simp` normal form." Lean developers mark a theorem with `@[simp]` when they want the `simp` tactic to apply it automatically. For example, the `mul_one` theorem is marked with `@[simp]` since, in most cases, $x * 1$ should be replaced with just x automatically.

However, when only a particular subset of lemmas should be applied, one can instead tell `simp` to use a library of lemmas marked with a label other than `@[simp]`. Doing so leads to a faster simplifier that doesn't simplify proof terms "too far."

In this project, we define a new attribute called `chev_simps`, and we often invoke `simp only [chev_simps]` in our proofs. Many relations, such as the linearity, identity, and inverse relations, should be applied automatically in almost all cases to simplify an expression to normal form, and so these relations are marked with `chev_simps`.

We define a similar attribute `height_simps` for theorems involving the graded `height` function, and a tactic `ht` to discharge inequalities about heights (with `ht` basically being a wrapper for "`simp only [height_simps]; omega`").

Our second form of automation is theorem macros. In addition to the linearity relations, there are many theorems that can easily be derived from others. For example,

$$[\{\zeta, i, t\}, \{\eta, j, u\}] = 1 \iff \{\zeta, i, t\} \cdot \{\eta, j, u\} = \{\eta, j, u\} \cdot \{\zeta, i, t\}.$$

In other words, proving the relation on the left gives an easy way to derive the equation on the right. In Lean, these kinds of proofs are typically between three and ten lines, and they happen often enough in our formalization that we wrote macros that expand into an entire theorem statement and proof.

The simplest of these macros is the three for the linearity, identity, and inverse relations. Given a root system `w`, a ring/field `R`, and a root `r`, the macro roughly looks like this (with many Lean-specific meta-programming details omitted):

```
macro "declare_lin_id_inv_thms" w R r : command => do return `(
  @[simp, chev_simps]
  theorem $linOf_r : lin_of_root(($w $R).project, $r) :=
    PartialChevalley.lin_helper ($w $R) (by unfold $w; simp)

  @[simp, chev_simps]
  theorem $idOf_r : id_of_root(($w $R).project, $r) :=
    id_of_lin_of_root $linOf_r

  @[simp, chev_simps]
  theorem $invOf_r : inv_of_root(($w $R).project, $r) :=
    inv_of_lin_of_root $linOf_r
)
```

Basically, the parameters `w`, `R`, and `r` are passed to the correct definitions, and new theorems are declared with the names stored in the identifiers `$linOf_r`, `$idOf_r`, and `$invOf_r` (which are constructed by omitted code). For example, if the root is α, then the corresponding identity theorem would be named `id_of_`α.

We have similar macros to prove many different types of theorems, including the trivial commutator example above and the reflected theorems from Section 5. For example, we invoke the reflected theorem macro by giving it the reflected heights, like this:

```
declare_B3Large_triv_comm_reflected_thm F α α2β2ψ heights 0 5 to 1 0
declare_B3Large_triv_comm_reflected_thm F α α2β2ψ heights 1 1 to 0 4
declare_B3Large_triv_comm_reflected_thm F α α2β2ψ heights 1 2 to 0 3
```

Overall, our theorem macros automatically fill in many of the relations we get for free in the weak presentation, and they help reduce the amount of boilerplate code.

Our final form of automation is group reassociation. Consider the theorem `mul_comm`, which states that any two elements commute under a commutative operation $*$: $a * b = b * a$. Now suppose we want to prove that $c * a * b = c * b * a$. Showing this seems simple: merely apply `mul_comm a b`. However, multiplication in Lean is left-associative by default, and so the application of `mul_comm a b` fails, since the term actually looks like $(c * a) * b = (c * b) * a$. In this case, we would need to manually reassociate the terms to look like $c * (a * b) = c * (b * a)$ before we could apply `mul_comm a b` and finish the proof.

However, we want to avoid doing this kind of manual reassociation in our formalization, since our theorems often involve group equations involving products of five or more terms. Thus, we modified a trick from `mathlib`'s category theory library, `reassoc_of%`,[7] to apply to groups. In the category theory setting, `reassoc_of%` reassociates an arrow to the right by introducing a new term on the left. In our case, we wrote `greassoc_of%` ("group re-associate") to multiply by a new group element on the left. Informally, it does the following:

`greassoc_of%` $(a * b) \coloneqq \forall c,\ (c * a) * b = (c * b) * a.$

Thus, `rw [greassoc_of% (mul_comm a b)]` might succeed where `rw [mul_comm a b]` fails.

To spare us from typing `greassoc_of%` each time we wanted to use it, we developed a `grw` tactic ("group rewrite") that behaves very similarly to `rw`. For each rewrite rule ℓ given to the tactic, `grw` first tries to apply ℓ. If that fails, it then tries to apply `greassoc_of%` ℓ. After each successful rewrite, the tactic applies a set of lemmas very similar to `chev_simp` in an effort to simplify the proof term. This automation sped up our proof efforts considerably, and it also made the proofs faster to compile and easier to read and maintain.

Unfortunately, `grw` is too tailored to our project to be broadly useful, but a general tactic could be teased from it with enough tinkering.

7 Conclusion

In this work, we used the Lean theorem prover to verify pen-and-paper calculations showing that a subset of the Steinberg relations defining three specific Chevalley groups imply the full set. To the best of our knowledge, our work is the first computer verification of these kinds of results.

Throughout our project, Lean was both a help and a hindrance: it checked our calculations with machine accuracy, but its quirks made it difficult to perform those calculations. Our custom automation eased this problem somewhat, but there is plenty of room for improvement.

[7] Simon Hudon wrote the version of this macro for Lean 3. The version in Lean 4 is mainly due to Kim Morrison. See its source file in mathlib4 on GitHub for more information.

There are numerous exciting directions which follow from our work. These include:

- Formalizing presentations for the A_n graded unipotent group for general n, which was also studied in [25]. This might be interesting because the positive root system is no longer of a fixed size, and thus a more-general argument will be needed.
- Automating the generation of the group-rewriting proofs. Most of these proofs involve only a handful of relations and general facts on groups and fields. While group rewriting problems are undecidable in general, a custom proof search procedure could exhaustively look for proofs on a given set of lemmas and a fixed number of rewrites to try.
- Formalizing presentations for the B_4 or C_3 graded unipotent groups. To the best of our knowledge, these have not been studied in any works. We hope that our formalization, however arduous, makes future calculations with these groups much easier – both by guiding those done by hand, and by smoothing the path of those in a future formalization. We are also interested in how to automatically generate proofs for these groups.
- Delving further into the land of Chevalley groups and root systems. This could include formalizing diagonal relations; formalizing recent work [26] that studies presentations of the full, non-unipotent type-A_n Chevalley groups; or formalizing the notion of additive structure in a (positive) root system and integrating this with the existing root system functionality in `mathlib`. Accomplishing the latter would likely make certain parts of our current formalization much easier, such as defining the lifting homomorphisms.
- Formalizing the notion of the *Dehn function* in Lean. In combinatorial group theory, the Dehn function measures the "complexity" required to derive one equation from others. In particular, the Dehn function of a purported equation in a presentation is finite iff the equation holds in the presentation, i.e., if it can be derived from the starting set of relations. To make their quantitative claims about HDX, KOOS [25, 35] need a stronger statement than we formalize: They need the Dehn function of the full set of relations in the graded group, relative to the weak set of relations, to be bounded independently of q (the size of the underlying field). However, their proof is somewhat unsatisfactory: they look at the derivations they have already found and simply observe that nowhere is there dependence on the underlying field. Formalizing Dehn functions seems quite tantalizing from the perspective of `mathlib`, and designing accompanying tactics to do rewrites while keeping track of the accumulating length of the derivation would be an interesting opportunity to peer further into the underlying mechanics of Lean.

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